Metaplectic Jacquet–Langlands Transfer, Companion Paper (Rank Two, Wild Place)
Revision Log: Draft 3 to Draft 4

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November 8, 2026

Scope

Draft 4 makes exactly the six corrections requested in ChatGPT's referee report on Draft 3, with no change of mathematical scope. All six were checked and found correct before being applied. Sections other than 8.2 (one lemma statement), 12.1, 13 are unchanged.

The six corrections

1. **Central character in the two-group step** (Theorem `thm:reduction-all`). Draft 3 twisted $\rho^\vee$ by $\theta_1^2 \circ \mathrm{Nrd}$ and called the result trivial on $F^\times$ (even case) or of central character sgn (odd case). Since $\mathrm{Nrd}(z) = z^2$, that twist restricts to θ_1 on scalars, so both claims were wrong. Following the referee, the proof is now organized around the genuine order-three character ψ_P of Γ_{P_0} (trivial on a Sylow 2-subgroup, $\psi_P(z,0) = \theta_1(z)$, hence $\psi_P(x_\eta) = \theta_1(\eta^4)$) and the untwisted $\rho^\vee$:

   $$\phi_\rho|_{\Gamma_{P_0}} = \psi_P \otimes \Big(3\rho^\vee|_{D_8} + 3\dim\rho \sum_\pi (\dim\pi)\Theta_\pi\Big) \quad \text{resp.} \quad \psi_P \otimes (3f\,\Theta_{\rho^\vee|_{SD_{16}}} + 3\dim\rho\,(2\chi_1 + 2\chi_2)).$$

   The SD_{16} identification and the swap $f\chi_1 = \chi_2$ are unchanged (the referee verified them).

2. **Brauer case split** (same theorem). When $T \supset \mu_2$ a noncyclic B_2 can have cyclic image in D_8 (e.g. $\langle a^4, b\rangle$), and $C_{\Gamma_0}(B_2)$ then contains noncentral elements of the j-torus, so Draft 3's "noncyclic B_2" case was wrong. The split is now by the image $\bar{B}_2 \subset D_8$: trivial (hypothesis on Φ_ρ), nontrivial cyclic (toral descent, using $B_2 \subset \langle y\rangle(B_2 \cap T)$ and $T \subset T_{E_y}$), noncyclic (centralizer lemma, then the two-group computation). Lemma `lem:centralizers`(b) is restated for 2-subgroups with noncyclic image, which is what its proof always used; for $\rho = \mathbf{1}$ ($T = \mu_3$) the two formulations agree, so Theorem `thm:reduction` is unaffected.

3. **General-p character formula** (Theorem `thm:H1-gen`). The value of ψ_{2d} at the star lift is $\theta_0(z)$ with $z \equiv \mathrm{Nrd}(X)^{(p+1)/2}$, so the formula is $\zeta^{e-\kappa(X)}\theta_0(\mathrm{Nrd}X)^{(p+1)/2}$, with $\theta_0^p = 1$ now stated; Draft 3's exponent -1 is correct only for $p = 3$. The multiplicativity proof is unchanged.

4. **Depth lemma** (Lemma `lem:depth-gen`(c)). Both cases are now written out: ramified via the isomorphism $U_F^{(d)}/U_F^{(d+1)} \to U_E^{(2d)}/U_E^{(2d+1)}$ and bijectivity of the p-th power map $U_E^{(k_0+1)} \to U_E^{(2d+1)}$; unramified via the boundary map $\bar{a} \mapsto \bar{a}_0^p \wp(\bar{a}/\bar{a}_0)$ (with $p/\varpi^e \equiv -\bar{a}_0^{p-1}$ from $\zeta_p^p = 1$), whose image is $\bar{a}_0^p \ker \mathrm{Tr}_{k_E/\mathbb{F}_p}$ and misses k_F because $\mathrm{Tr}_{k_E/\mathbb{F}_p} = 2\,\mathrm{Tr}_{k_F/\mathbb{F}_p}$ on k_F and p is odd.

5. **Typo** in Theorem `thm:levelone`: $w^4 = \mathrm{Nm}(w)$, not $\mathrm{Nm}(w)^2$.

6. **Citation and summation by parts.** Lemma `lem:model`(a) now cites Kubota's 1967 paper for SL_2 and identifies the GL_2 formula as the $r = 2$, $c = 0$ Kazhdan–Patterson cocycle with the n-th power symbol (added `[Kub67]` to the bibliography). The summation by parts in Theorem `thm:pthpower` is written out: $c_{j-1} - 2c_j + c_{j+1} = -1$, $c_1 = c_p = 0$, $c_0 = c_{p+1} = -p/2 \equiv 0$, and the four boundary terms are displayed.

Status

Theorem `thm:reduction-all` [P] (repaired); Theorem `thm:H1-gen` [P] (formula corrected, lemma completed); Theorem `thm:pthpower` [P] (accepted by the referee); Theorem `thm:levelone` [P^c] (conditional on Target A, as before). Nothing else changed status.